PARENTAL EARNINGS AND CHILDREN'S WELL-BEING AND FUTURE SUCCESS:

AN ANALYSIS OF THE SIPP MATCHED TO SSA EARNINGS DATA

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Abstract

We estimate the association between parental earnings and a wide variety of indicators of child well-being using data from the Survey of Income and Program Participation (SIPP) matched to administrative earnings records from the Social Security Administration. We find that the use of longer time averages of parent earnings leads to substantially higher estimated effects compared to using only a single year of parent earnings. This suggests that previous studies may have understated the potential efficacy of income support programs to improve child well-being. Further, policy makers should take into account the attenuation bias when comparing studies that use different time spans to measure parental income. Using 7 year time averages of parent earnings, we show for example, that a doubling of parent earnings reduces the probability of a teenager reporting being in poor health by close to 50 percent and a child having insufficient food by 75 percent.

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1. Introduction

In the aftermath of the Great Recession, approximately one in five children in the United States lives in poverty, making children the poorest age group in the United States (Children's Defense Fund 2010, Land 2010). Comprehensive measures of child well-being have also begun to decline after showing steady improvement for most of the past 20 years. A growing literature across many disciplines has emphasized that the conditions faced by children in the first few years of life may be especially important for long-run socioeconomic success (e.g., Almond and Currie, 2011).

Given this backdrop, efforts to improve the material conditions of children are likely to remain a salient issue for policymakers for many years. An obvious target for policy makers is to improve the economic situation of parents. Indeed, a vast literature in the social sciences has studied the association between parental income and children's outcomes to establish the importance of parental economic resources on children's well-being. One limitation of most of these studies is the lack of availability of parental earnings histories over long periods of time for a very large and representative sample of families in the US. It is well established that the bias in using single year measures of parental income to proxy for long-run income can be sizable and can vary by parental age (Solon, 1992; Mazumder, 2005; Haider and Solon, 2006). Therefore the findings of previous studies may have sharply underestimated the role of parental resources in determining children's health, education and future socioeconomic success. Policymakers, therefore, may have underestimated the potential efficacy of income support policies and may have misperceived the appropriate mix of policy interventions that are required to improve children's outcomes. This paper directly addresses this issue by using a rich dataset that includes a wide variety of measures of child well-being linked to administrative data on parental earnings taken over many years. This allows us to assess the degree of attenuation bias from using only a single year, or just a few years of earnings.

Our data set pools families from the 1984, 1990-1993, 1996, 2001 and 2004 panels of the Survey of Income and Program Participation (SIPP). Each of these SIPP samples were matched to earnings histories contained in SSA administrative earning records. We use the administrative data to construct long-term time averages of parents' earnings. We use these time averages of parent earnings to estimate the association between parent earnings and childhood well-being. We use SIPP topical modules on Children's Well-Being, Functional Limitations and Disability, Health Status and Utilization of Health Care and Extended Measures of Well-Being to obtain a broad set of measures related to childhood health and well-being.

Our results show that across virtually all of our indicators of childhood well-being, longer time averages lead to substantially larger estimates. In some cases the effects more than double compared to using a single year of earnings. In certain instances the estimates are only statistically significant when using the longer time averages. This provides strongly suggestive evidence that previous results of no significant findings or small effect sizes may have been spurious and should be reconsidered in light of the evidence in this study.

Among our substantive findings we show that a doubling of a 7-year average of earnings is associated with a 48 percent drop in the mean probability of a teenager reporting poor health, a 39 percent drop in the mean probability of repeating a grade, an 18 percent drop in the mean probability of being suspended, a 75 percentage point drop in the mean probability of having insufficient food and a 6 percentage point increase in the mean number of times a child saw a doctor. Notably, these effects are 25 to 120 percent larger than the effects estimated using a single year of earnings.

While we think that our study provides an important contribution, an even bigger remaining challenge is to convincingly demonstrate that these statistical associations between parental earnings and children's outcomes, even if better estimated, actually reflect causal processes. Developing research designs that convincingly address this question is a difficult endeavor and is not our aim in this paper.

Instead like most of the preceding literature linking parental income to children's well-being, we hope to provide descriptive estimates that may provide important insight for policy discussions and help inform future research on child well-being. To the extent that our results do reflect a causal relationship, however, they could have important implications for income support policies, such as the Earned Income Tax Credit, TANF and Food Stamps.[1] Moreover, the possibility of severe attenuation bias due to measurement error in income should be considered by policy makers when evaluating previous studies linking parental resources to child well-being.

2. Background

There is an enormous literature that discusses the many potential determinants of childhood well-being. This paper focuses on the role of just one of these factors, parental labor market earnings.[2] It has been well established that parent income is clearly amongst the most important if not the most important determinant of child well-being (Brooks-Gunn and Duncan, 1997). Therefore, it is critical that parent income be well measured and its effects estimated as accurately as possible.

Issues related to the measurement of parental income have played an important role in the development of the literature on intergenerational economic mobility. Researchers have typically estimated a regression of children's log income on parent's log income. The regression coefficient also known as the intergenerational elasticity, measures the degree of persistence in income and one minus this coefficient has been used to infer the degree of intergenerational mobility. The first set of estimates of this regression typically used only single year measures of income in each generation producing estimates of the intergenerational elasticity of income of around 0.2. This suggested that there was substantial mobility in the U.S. and that on average income differences between families would be wiped out within three generations (Becker and Tomes, 1986).

[1] Hoynes and Schanzenbach (2009) show that many households respond similarly to a dollar in food stamps as they do to a dollar in income.

[2] Although we use income and earnings interchangeably in this section for ease of exposition, all of the empirical estimates concern labor market earnings.

The central idea that changed the consensus view of intergenerational mobility stems from Milton Friedman's (1957) insight that economic behavior is more strongly related to permanent income than transitory income. Bowles (1972) was perhaps the first to apply this idea in the context of intergenerational mobility by suggesting that the transitory income fluctuations could bias down estimates of the degree of intergenerational persistence that were based on just using a single year of income. Solon (1989, 1992) demonstrated the bias more formally and showed how even short multi-year averages of income could dramatically reduce such bias. Solon's (1992) study substantially revised the consensus view of the intergenerational elasticity in income from 0.2 to 0.4 or possibly higher.

Building on Solon's work, Mazumder (2005) argued that persistent transitory fluctuations could lead to non-negligible bias in even short term averages of parent income. Using both simulations and actual estimates based on a new intergenerational sample derived from the 1984 SIPP matched to social security earnings records, Mazumder argued that estimates of the intergenerational elasticity in the US may be as high as 0.6. This would suggest that earnings differences would take several more generations to be eliminated. Simulations from Mazumder (2005) also provide estimates of the reliability ratio for multi-year averages of income when the explanatory variable of interest is permanent income. These show that coefficients using a single year of parent income are biased down by about 50 percent and that even 5 year averages are biased down by about 30 percent.

For the most part, however, this key insight regarding the importance of averaging parental income over many years has not been utilized by social scientists for analyzing most other outcomes related to child well-being. One exception is a recent paper by Rothstein and Wozny (2009) who show that much more of the black-white test score gap can be explained when using long-term averages of parent income than using just one year. This substantially revises the estimates of Fryer and Levitt (2006) concerning the amount of the black-white test score gap that is unexplained. Similarly, Sullivan, Turner and Danziger (2008) using samples of low income women from the Women's Employment Study and

1996 SIPP show that the probability of experiencing a material hardship is more strongly correlated with permanent income than transitory income.

One important reason why researchers have not explored the role of long-term averages is simply due to data considerations. Outside of the PSID, few data sources contain panel data on parental income.[3] A limitation of the PSID is that sample sizes can be relatively small. This limitation is amplified if one is concerned about the representativeness of the sample due to ongoing attrition since the beginning of the sample in 1968.[4]

A more fundamental issue is whether the explanatory variable of interest in a particular study *ought* to be permanent income or current income. In the case of intergenerational mobility, social scientists have been interested in a measure of income that spans the lifetime income of each generation so permanent income is the natural concept to use. In other cases, the choice may be driven by theoretical considerations and may depend to some degree on the plausibility of the existence of borrowing constraints. In the case of understanding the importance of parental income on childhood well-being there is a strong case to be made that the object of interest is income received during the childhood of the child (perhaps under the assumption that borrowing constraints exist for some households). In this case it may be optimal to use a short-time average (e.g., 5 to 10 years) of parent income both because we are interested in income during this period and because of the desirable properties of time averages in reducing bias. Given the potential important changes in interpretation of existing socioeconomic gaps or income gradients there is a strong case for at a minimum, exploring how estimates are altered when using multi-year averages of parental income.

[3] An exception is the Children of the NLSY, where income from the parents can be linked from the NLSY79.
[4] An additional issue is raised if one uses data from the SEO (oversample of poorer households) component of the survey where there were problems with the implementation of the initial sampling scheme (Brown, 1996).

3. Data and Methodology

The analysis uses the 1984, 1990, 1991, 1992, 1993, 1996, 2001 and 2004 panels of the Survey of Income and Program Participation (SIPP) matched to Social Security Administration's (SSA) Summary Earnings Records (SER) and Detailed Earnings Records (DER).[5] The SER data covers annual earnings over the period from 1951 to 2007, while the DER data is available from 1978 onward. There are two aspects to using SER records that raise potential issues. The first is that some individuals who are working are not covered by the social security system and their earnings will be recorded as zero. Second, earnings in the SER data are censored at the maximum level of earnings subject to the social security tax. While in principle the DER data is not subject to either of these problems, an examination of the data shows that the DER data actually shows higher rates of non-coverage than the SER data. Since the non-coverage patterns are different in the two datasets, we take the maximum of earnings in a year between the SER and DER to minimize the bias due to non-coverage. To deal with the top-coding problem in the SER we impute earnings among the topcoded for each year starting in 1961 by using the March CPS and calculating the mean earnings among those above the topcode by cells defined by race, sex and education.

The sample consists of children who were 0 to 20 years old and who were co-resident with at least one parent at the time of their first interview. The sample was also restricted to children whose parents were between 15 and 45 years old when the child was born. We progressively average parent earnings over 1, 3, 5 and 7 years. The earnings of the father are used if his earnings were positive for all 7 of the years. If not, we use the mother's earnings if her earnings were positive in all 7 years. If neither had positive earnings in all 7 years, the family was dropped from the sample. The time period covered by the averages ends in the year prior to the interview in which the observation is used. For example, if an outcome was taken from an interview that occurred in October 1992, the one-year average would use

[5] The match rates in most years are quite high (around 80 to 90 percent) and an analysis of the match probability by demographic characteristics suggest that selection is not a major concern. Mazumder (2005) has previously shown that the match rate between the 1984 SIPP panel and the SER data is around 90 percent and that correcting for selection based on inverse probability weighting has little effect on intergenerational regressions.

1991 earnings, the three- year average would use 1989 through 1991 earnings, the five-year average would use 1987 through 1991 earnings and the seven-year average would use 1985 through 1991 earnings. Consequently, time averages were generated for each specific outcome depending on the SIPP panel used and the year of the interview.

We organize the outcomes into five distinct groups. Summary statistics are shown in Table 1. The first group is a set of general health outcomes. This includes: 1) an indicator for a physical, learning or mental condition that limits schoolwork which is asked of children 5 or older; 2) an indicator for a physical, learning or mental condition that limits child behavior asked of children younger than 5; 3) an indicator for poor health among children 15 and older that uses health status (based on health status rated on a 1 to 5 scale where 1 is excellent and 5 is poor) reported in the Functional Limitations and Disability Topical Module (child response); 4) an indicator for poor health that uses health status reported in the Children's Well-Being Topical Module (parent response); 5) an indicator for poor health that combines health status reported in the Children's Well-Being Topical Module and health status reported in the Functional Limitations and Disability Topical module; 6) an indicator for spending the night in a hospital in the last year; 7) the number of nights spent in the hospital in the last year; and 8) the number of days in the last four months that illness or injury kept the individual in bed for at least half the day.

The second set of outcomes deal with health care utilization and include: 1) the number of times the child talked to a doctor in the last year; 2) the number of dentist visits in the last year and 3) an indicator for using prescription drugs daily. The third group of outcomes is also health related and includes three anthropomorphic measures of children below the age of 5. These include: 1) a weight-for-height Z-score; 2) a weight-for-age Z-score; and 3) a height-for-age Z-score. All three measures are calculated using 2 or 3-month age bins separately for males and females.

The fourth group of outcomes examines a range of childhood educational measures. These include: 1) the number of times the child changed schools; 2) an indicator for having repeated a grade; 3)

an indicator for having been suspended or expelled; 4) an indicator for having received special education services; and 5) an indicator for having a learning disability.

The fifth and final set of outcomes examines a range of measures related to home environment and family resources. These include: 1) the number of times a child was read stories in the last week; 2) a count of the number of days without enough food or money to buy food in the last month; 3) an indicator for whether the child had ever been in day care; 4) an indicator for not being able to meet basic needs (food, rent, utilities, etc.) at some point in the last year; 5) an indicator for the family not having enough food in the last four months; and finally 6) an indicator for a family member skipping a doctor visit when he or she needed to go. The last four outcomes are estimated using one observation per family.

All of the outcomes (denoted by y_i) are multiplied by 100 for convenience in displaying and interpreting results. This is mostly useful for the indicator variables so that the coefficients can be interpreted as percentage point effects. For each regression, we include a basic set of covariates ("Basic Controls") which consist of indicators for survey year, child age when the outcome was measured, race, ethnicity, gender, state of residence, having a one parent household headed by the mother, having a one parent household headed by the father and an indicator for using father's earnings. A more extensive set of controls ("Added Controls") includes all of the basic controls and adds parent education, parent health status, parent age and parent age squared. For the regressions dealing with health we also added a third set of controls which include separate indicators for the father, mother and child having private health insurance. These controls are contained in the vector X_i. The results are weighted using the child's person weight in the fourth month of the wave in which the outcome was observed. The main object of interest is γ in the equation below:

$$(1) \quad y_i = \gamma Inc_i + \beta X_i + \varepsilon_i .$$

4. Results

We start by presenting the results concerning children's health outcomes in Table 2. The first entry in the table, -0.979, shows the coefficient on using log parental earnings from a single year where the outcome is an indicator for having a health condition that prevents school work. The standard error is 0.15 and the effect is significant at the 1 percent level. The point estimate suggests that a doubling of parental earnings would lead to a reduction in the probability of such a health condition by approximately 0.7 percentage points.[6] Since the mean rate of such health conditions is about 5 percent, this would be about a 14 percent effect size evaluated at the mean. This would essentially be what the typical researcher would estimate when using only the current income available in the survey. Moving across the row the next three columns shows how the effect changes as the length of the time average is increased. In column (4), when we use a 7 year time average, the coefficient rises (in absolute value) to -1.37. Moving from a single year of parent earnings to a 7 year average raises the coefficient by 40 percent. The estimate based on a 7-year average implies that a doubling of earnings reduces the probability of a schoolwork limiting health condition by about 0.95 percentage points.

Columns (5) through (8) use the same lengths of time averages but now include the added set of covariates on parental characteristics. This sharply reduces the point estimates. For example, when using a 7 year average (column 8), a doubling of parental earnings now reduces the probability of a health condition by 0.77 percentage points. The difference between using a single year and a seven year average, however, remains substantial. In fact, the coefficient on using a seven year average is now almost 50 percent higher than the coefficient on a single year of earnings. Finally, in column (9) we continue to use a 7 year average but now include a set of indicators for health insurance coverage. This further reduces the effect of doubling parent earnings to -0.61 percentage points. Whether one wants to

[6] Since our right hand side variable is in logs in order to calculate the effect of a 100 percent increase we multiply our coefficient by ln(2) which is approximately 0.69. We have done this throughout our exposition of the results. Alternatively, to estimate the effect of say, a 10 percent increase in income we would multiply our coefficient by ln(1.1) or 0.09.

condition on additional covariates, such as parent age, health and education or health insurance coverage depends on the question of interest. Blau (1999) argues that if one is interested in identifying the reduced form effect of income, then covariates that are potentially jointly chosen with inputs to the child production function like parent education and parent age at birth should be excluded. Column (4) would be the preferred specification in this case. Alternatively, Brooks-Gunn and Duncan (1997) and many others prefer a strategy that adjusts for family characteristics that are potentially correlated with income, presumably under the assumption that these are not endogenous to income. Columns (8) and (9) adjust for many of these other characteristics. Since one reason why parental income might matter for health is precisely because it enables one to access health insurance, it may be of interest to know the full effect of income unconditional on insurance coverage. For that reason, one might choose column (8) as a preferred specification, though the results in column (9) are certainly of interest and are suggestive of the potential role of health insurance access.

The second row shows the analogous effects on the probability of a health condition that affects the behaviors of children at or below the age of 5. Here the incidence rates are much lower and so we might expect to find point estimates that are smaller in absolute value than the limitations on school work. For this outcome, however, we see no gradient in the time average of parent earnings and the reverse pattern of smaller estimates corresponding to longer time averages when we include controls. This turns out to be the only outcome in Table 2 where we find a reversal of the pattern.

We next turn to the outcome of poor self-reported health status among teenagers. The results in column (4) using a seven year time average (-1.74) imply that a doubling of parent earnings would reduce the probability of poor health among children by 1.2 percentage points, an effect that is significant at the 1 percent level. The coefficient from using only a single year of earnings is less than half the size. When additional controls are added, the results using the 7 year average drop by about 25% but remain highly statistically significant. In contrast, the estimate from using a single year of earnings is no longer significant when additional controls are included. Since many previous studies that examine the

relationship between income and child outcomes adjust for these additional outcomes and use single year income or short averages, one should question whether previous findings may have been biased towards zero because of measurement error.

When the outcome is parent reported health status for a child younger than 15 (row 4) the coefficients are substantially smaller in absolute value and much less robust to the inclusion of additional controls. As one would expect, when the two health status outcomes are combined to form one outcome (row 5), the effects are generally close to a weighted average of the individual results.

For the probability of staying overnight in a hospital, the results are fairly robust. All of the coefficients on the time averages of parent earnings are negative and significant, at a minimum, at the 10 percent level. The results are similar even when controlling for covariates (columns 5 to 8) and even when we include health insurance status (column 9). A doubling of parent earnings reduces the probability that a child will stay overnight in a hospital by about 0.3 percentage points and implies an effect size of about 10 percent evaluated at the mean. If we focus on column (8), we also find negative coefficients that are significant at the 10 and 5 percent level, respectively, for the number of days spent in a hospital and the number of days that illness kept the child in bed at least half the day. The effect sizes for these outcomes are 17 and 9 percent, respectively.

In Table 3, we turn to health care utilization outcomes. We find that for all three outcomes using a 7-year average leads to substantially higher point estimates on the effect of parental earnings than using a single year. When only the basic controls are included, the coefficients increase from between 37 percent to 42 percent. Notably for the number of dentist visits and daily prescription drug use, controlling for parent characteristics reduces, but does not eliminate, the effects which are all generally highly statistically significant. Using the column (8) results, the effect sizes for these outcomes, evaluated at the mean, range from 2 percent to 10 percent.

In Table 4, we examine the effects of parental earnings on the height and weight of children under the age of 5. The first outcome examined is the z-score of weight for height which is sometimes used to classify "wasting" or malnutrition for low levels. Of course, high levels can also be indicative of potential problems such as Type II diabetes. Regardless of whether the basic or added controls are included, we find a small negative effect of between 0.02 and 0.04 standard deviations from a unit increase in log parental earnings that is statistically insignificant. Similarly no consistent, let alone, statistically significant effect is found for weight for age or height for age. Interestingly, the effect of earnings on height for age becomes smaller as more years of earnings are included. This could be evidence that transitory income in the first year or two of life is important for height but that permanent income is not.

The fourth set of measures of child well-being deal with educational outcomes and the results are presented in Table 5. We find a negative effect on the number of school changes that rises with the length of the parental earnings time average. Using a 7-year average with the baseline and additional controls yields a marginally significant effect. The imprecisely estimated coefficients are suggestive of an effect size of about 7 or 8 percent evaluated at the mean. The effects are highly robust for ever repeating a grade or receiving special education services. The specification in column (8) suggests that doubling parent earnings reduces the likelihood of repeating a grade by 1.9 percentage points (25 percent effect size), reduces suspensions by 0.8 percentage points (9 percent effect size) and reduces placement in special education classes by 1 percentage points (11 percent effect size). Further, these results are a powerful example of how lengthening the time averages of parent earnings can dramatically alter the size of the estimated effects. We find that the estimated coefficients are between 30 and 230 percent larger when using a 7-year average than when using current year earnings. For learning disability, we find that some statistically insignificant earnings effects appear when using only the baseline controls but that these are completely removed when controls for parental characteristics are added.

The estimates for the final group of measures on home environment and family resources are shown in Table 6. The results are mixed. We find that for two of the outcomes, number of times read stories and days without food, there are highly significant effects when we use the baseline controls that are dramatically reduced and no longer significant once we further control for parental characteristics. However, for the other outcomes, ever been in day care, inability to meet basic needs, food inadequacy and skipped doctor visits, the results are highly robust. The implied effect sizes for these outcomes are 4 percent, 36 percent, 45percent and 50 percent, respectively.

While it may seem obvious that parental earnings will be strongly associated with two of the outcomes (inability to meet basic needs and food inadequacy) since these outcomes essentially reflect the availability of parental resources, it is important to point out that once again, the 7 year time averages yield significantly higher coefficients than current earnings. For example for food inadequacy, the effect size is nearly 10 to 25 percent higher when using a 7-year average than when using current earnings. So even for these most basic indicators of well-being it is clear that the longer time averages matter.

5. Conclusion

This analysis constructs a unique dataset linking a wide range of child well-being measures to administrative earnings histories of parents. This data enables us to better estimate the association between parental earnings to indicators of child well-being than most previous studies. By assembling long earnings histories for parents we reduce the attenuation bias that arises from using only current year income as an explanatory variable. For several critical outcomes we show that 7-year averages of parent earnings lead to estimates of effect sizes that are substantially higher than are obtained using only a single year of earnings data. We further show that some (but not all) of these outcomes are also robust to including a rich set of covariates including parental characteristics that are often absent in many cross-sectional datasets.

Our results have several potential implications for policy makers. The robust empirical finding across a variety of child outcomes that shows that the use of current earnings can attenuate the estimated associations should be taken in to consideration when an analysis that uses a single year of income or a short average of income finds that a policy does not have a significant effect. For example, Rothstein and Wozny (2009) show that Fryer and Levitt's (2006) finding that the Black-White test score gap is largely unrelated to income differences is due to their limited time span of income received during childhood. Rothstein and Wozny find substantially larger effects when using longer time averages. Similarly, policy makers should revise the magnitude of the estimated associations between parent income and child well-being from prior studies. We also provide better estimates of the relationship between different measures of childhood well-being and parental earnings that are free of this attenuation bias. To the extent that the associations we document are causal, our results can be used to identify areas of child well-being that might be particularly responsive to different income support policies.

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Table 1: Summary Statistics of Measures of Child Well-Being

Outcome	N	Mean	SD
Health Outcomes			
Condition Limits School Work	48131	0.05	0.23
Condition Limits Child Behavior	6071	0.01	0.12
Health Poor Functional Limitations	6877	0.02	0.15
Health Poor Child Well Being	9569	0.02	0.14
Poor Health	15318	0.02	0.15
Night in Hospital	27280	0.03	0.16
Number of Nights in Hospital	27280	0.14	1.88
Number of Sick Days Last 4 Months	27280	0.65	4.67
Health care use			
Number of Times Talked to Doctor in Last Year	27280	2.65	6.07
Number of Trips to the Dentist	25061	1.62	2.47
Daily Prescription Drug Use Last Year	21856	0.11	0.31
Physical Characteristics			
Weight-For-Height z-score	2058	-0.06	0.93
Weight-For-Age z-score	2058	-0.02	0.96
Height-For-Age z-score	2058	-0.03	0.99
Educational Outcomes			
Number of Times Changed Schools	9732	0.43	0.98
Ever Repeated a Grade	9732	0.07	0.26
Ever Been Suspended	4655	0.09	0.29
Special Education	41592	0.09	0.28
Learning Disability	40641	0.02	0.14
Home Environment and Family Resources			
Number of Times Read Stories	5862	5.31	6.17
Days Without Food or Food Money	3897	0.16	2.04
Ever Been in Daycare	10606	0.38	0.49
Was Unable to Meet Needs in Last Year	3897	0.17	0.37
Not Enough Food in Last 4 Months	3897	0.02	0.14
Someone in Family Skipped Doctor	3897	0.08	0.27

Table 2: Effects of Parental Earnings on Children's Health

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Basic Controls				*Added Controls*				*with*
	Parent earnings averaged over…				*Parent earnings averaged over…*				*Hlth Insur.*
*Outcome*100*	*1 yr*	*3 yrs*	*5 yrs*	*7yrs*	*1 yr*	*3 yrs*	*5 yrs*	*7yrs*	*7yrs*
Condition that limits	-0.979	-1.259	-1.331	-1.374	-0.757	-1.01	-1.068	-1.114	-0.892
school work	[0.150]***	[0.182]***	[0.195]***	[0.198]***	[0.160]***	[0.199]***	[0.216]***	[0.220]***	[0.221]***
	48131	48131	48131	48131	48131	48131	48131	48131	48131
Condition that limits	-0.429	-0.441	-0.439	-0.39	-0.319	-0.251	-0.215	-0.155	-0.152
child behavior	[0.216]**	[0.233]*	[0.258]*	[0.266]	[0.232]	[0.263]	[0.298]	[0.326]	[0.350]
	6071	6071	6071	6071	6071	6071	6071	6071	6071
Poor Health Status	-0.801	-1.354	-1.566	-1.741	-0.376	-0.883	-1.075	-1.271	-0.985
(Func. Lim. Module)	[0.300]***	[0.363]***	[0.401]***	[0.404]***	[0.340]	[0.403]**	[0.446]**	[0.432]***	[0.422]**
	6877	6877	6877	6877	6877	6877	6877	6877	6877
Poor Health Status	-0.583	-0.674	-0.695	-0.745	-0.35	-0.368	-0.357	-0.409	-0.062
(Child WB Module)	[0.277]**	[0.289]**	[0.302]**	[0.304]**	[0.300]	[0.326]	[0.351]	[0.363]	[0.372]
	9569	9569	9569	9569	9569	9569	9569	9569	9569
Poor Health Status	-0.605	-0.917	-1.015	-1.127	-0.251	-0.509	-0.582	-0.707	-0.488
(Pooled)	[0.201]***	[0.238]***	[0.255]***	[0.255]***	[0.227]	[0.269]*	[0.291]**	[0.287]**	[0.289]*
	15318	15318	15318	15318	15318	15318	15318	15318	15318
Stayed in Hospital	-0.23	-0.347	-0.405	-0.418	-0.188	-0.327	-0.399	-0.417	-0.344
	[0.120]*	[0.148]**	[0.158]**	[0.159]***	[0.129]	[0.165]**	[0.180]**	[0.184]**	[0.189]*
	27280	27280	27280	27280	27280	27280	27280	27280	27280
# Nights in Hospital	-1.731	-2.05	-2.02	-2.162	-2.204	-2.933	-3.07	-3.431	-2.367
	[1.061]	[1.393]	[1.505]	[1.729]	[1.007]**	[1.316]**	[1.517]**	[1.841]*	[1.626]
	27280	27280	27280	27280	27280	27280	27280	27280	27280
# of Sick Days	-2.275	-5.228	-6.215	-7.632	-0.837	-4.762	-6.285	-8.378	-8.827
	[2.391]	[3.051]*	[3.451]*	[3.874]**	[2.437]	[3.043]	[3.578]*	[4.203]**	[4.561]*
	27280	27280	27280	27280	27280	27280	27280	27280	27280

Table 3: Effects of Parental Earnings on Health Care Use

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Basic Controls				*Added Controls*				*with*
	Parent earnings averaged over…				*Parent earnings averaged over…*				*Hlth Insur.*
*Outcome*100*	*1 yr*	*3 yrs*	*5 yrs*	*7yrs*	*1 yr*	*3 yrs*	*5 yrs*	*7yrs*	*7yrs*
Frequency	16.495	20.162	21.29	22.571	8.581	7.913	7.544	8.394	2.615
Talked to Doctor	[4.652]***	[5.582]***	[5.983]***	[6.206]***	[5.133]*	[6.740]	[7.473]	[7.827]	[8.393]
	27280	27280	27280	27280	27280	27280	27280	27280	27280
Number of	23.627	30.445	32.587	33.565	17.489	22.731	24.368	25.224	16.213
Dentist Visits	[1.965]***	[2.492]***	[2.668]***	[2.728]***	[2.108]***	[2.810]***	[3.095]***	[3.218]***	[3.484]***
	25061	25061	25061	25061	25061	25061	25061	25061	25061
Daily Prescription	0.839	1.152	1.21	1.195	0.59	0.793	0.811	0.756	0.644
Drug Use	[0.279]***	[0.342]***	[0.363]***	[0.370]***	[0.302]*	[0.387]**	[0.419]*	[0.432]*	[0.456]
	21856	21856	21856	21856	21856	21856	21856	21856	21856

Table 4: Effects of Parental Earnings on Physical Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Basic Controls				*Added Controls*			
	Parent earnings averaged over…				*Parent earnings averaged over…*			
*Outcome*100*	*1 yr*	*3 yrs*	*5 yrs*	*7yrs*	*1 yr*	*3 yrs*	*5 yrs*	*7yrs*
Weight for Height	-2.649	-4.636	-4.122	-3.495	-2.128	-3.656	-2.785	-1.966
z-score	[2.573]	[3.093]	[3.196]	[3.215]	[2.699]	[3.476]	[3.725]	[3.836]
	2058	2058	2058	2058	2058	2058	2058	2058
Weight for Age	-0.788	-3.687	-3.705	-3.85	-1.221	-3.32	-2.955	-3.098
z-score	[2.238]	[3.080]	[3.175]	[3.254]	[2.540]	[3.580]	[3.854]	[4.066]
	2058	2058	2058	2058	2058	2058	2058	2058
Height for Age	4.814	3.789	2.893	1.986	2.867	2.276	1.26	0.179
z-score	[2.942]	[3.295]	[3.517]	[3.611]	[3.122]	[3.989]	[4.451]	[4.779]
	2058	2058	2058	2058	2058	2058	2058	2058

Table 5: Effects of Parental Earnings on Children's Educational Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Basic Controls				*Added Controls*				*with*
	Parent earnings averaged over…				*Parent earnings averaged over…*				*Hlth Insur.*
*Outcome*100*	*1 yr*	*3 yrs*	*5 yrs*	*7yrs*	*1 yr*	*3 yrs*	*5 yrs*	*7yrs*	*7yrs*
Number of Times	-2.38	-3.294	-3.446	-4.684	-2.791	-3.919	-3.947	-5.136	
Changed School	[1.849]	[2.178]	[2.265]	[2.391]*	[1.981]	[2.402]	[2.511]	[2.699]*	
	9732	9732	9732	9732	9732	9732	9732	9732	
Ever Repeated	-2.259	-3.384	-3.697	-3.953	-1.375	-2.264	-2.483	-2.719	
A Grade	[0.360]***	[0.431]***	[0.457]***	[0.470]***	[0.383]***	[0.459]***	[0.494]***	[0.514]***	
	9732	9732	9732	9732	9732	9732	9732	9732	
Ever Suspended	-1.173	-2.051	-2.267	-2.391	-0.348	-0.989	-1.113	-1.161	
or Expelled	[0.498]**	[0.613]***	[0.663]***	[0.680]***	[0.551]	[0.701]	[0.771]	[0.791]	
	4655	4655	4655	4655	4655	4655	4655	4655	
Special Education	-1.223	-1.573	-1.593	-1.576	-1.055	-1.455	-1.486	-1.474	-1.34
	[0.219]***	[0.251]***	[0.265]***	[0.272]***	[0.241]***	[0.278]***	[0.296]***	[0.305]***	[0.314]***
	41592	41592	41592	41592	41592	41592	41592	41592	41592
Learning disability	-0.138	-0.191	-0.229	-0.266	-0.056	-0.103	-0.143	-0.186	-0.145
	[0.107]	[0.134]	[0.144]	[0.147]*	[0.110]	[0.143]	[0.155]	[0.157]	[0.159]
	40641	40641	40641	40641	40641	40641	40641	40641	40641

Table 6: Effects of Parental Earnings on Home Environment and Family Resources

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Basic Controls				*Added Controls*			
	Parent earnings averaged over…				*Parent earnings averaged over…*			
*Outcome*100*	*1 yr*	*3 yrs*	*5 yrs*	*7yrs*	*1 yr*	*3 yrs*	*5 yrs*	*7yrs*
# of Times Read	27.986	45.14	48.19	48.316	11.308	23.256	24.525	24.827
Stories, Past Week	[11.259]**	[12.630]***	[14.500]***	[15.294]***	[12.489]	[14.967]	[18.280]	[20.108]
	5862	5862	5862	5862	5862	5862	5862	5862
Days Without Food	-11.354	-12.786	-12.631	-13.67	-5.597	-5.776	-4.821	-6.06
(or Money for Food)	[4.111]***	[3.803]***	[3.800]***	[3.793]***	[4.950]	[4.210]	[3.992]	[3.596]*
	3897	3897	3897	3897	3897	3897	3897	3897
Ever Been in	3.081	2.965	3.37	3.368	2.233	1.62	2.07	2.185
Day Care	[0.623]***	[0.833]***	[0.895]***	[0.924]***	[0.690]***	[0.922]*	[1.012]**	[1.067]**
	10606	10606	10606	10606	10606	10606	10606	10606
Unable to Meet	-8.874	-10.727	-11.159	-11.567	-7.015	-8.315	-8.531	-8.971
Needs in Past Year	[0.809]***	[0.918]***	[0.975]***	[0.999]***	[0.857]***	[1.010]***	[1.088]***	[1.132]***
	3897	3897	3897	3897	3897	3897	3897	3897
Did not Have Enough	-1.658	-2.006	-2.013	-2.084	-1.141	-1.319	-1.223	-1.252
Food	[0.379]***	[0.434]***	[0.416]***	[0.419]***	[0.413]***	[0.469]***	[0.442]***	[0.429]***
	3897	3897	3897	3897	3897	3897	3897	3897
Family Member	-5.401	-6.313	-6.498	-6.547	-4.717	-5.583	-5.799	-5.968
Skipped Dr. Visit	[0.651]***	[0.744]***	[0.759]***	[0.770]***	[0.707]***	[0.842]***	[0.878]***	[0.903]***
	3897	3897	3897	3897	3897	3897	3897	3897